



Peptic ulcer disease

- The term 'peptic ulcer' refers to an ulcer in the lower oesophagus, stomach or duodenum, in the jejunum after surgical anastomosis to the stomach or ,rarely, in the ileum adjacent to a Meckel's diverticulum.
- Ulcers in the stomach or duodenum may be acute or chronic; both penetrate the muscularis mucosa but the acute ulcer shows no evidence of fibrosis. Erosions do not penetrate the muscularis mucosae.

Gastric and duodenal ulcer (GU , DU).

- The prevalence of peptic ulcer (0.1–0.2%) is decreasing in many Western communities as a result of widespread use of Helicobacter pylori eradication therapy but it remains high in developing countries.
- The male-to-female ratio for duodenal ulcer varies from 5 : 1 to 2 : 1, while that for gastric ulcer is 2 : 1 or less. Chronic gastric ulcer is usually single %90 are situated on the lesser curve within the antrum or at the junction between body and antral mucosa .
- Chronic duodenal ulcer usually occurs in the first part of the duodenum and %50 are on the anterior wall. Gastric and duodenal ulcers coexist in 10% of patients and more than one peptic ulcer is found in 10–15% of patients.

Pathophysiology: H. pylori

Peptic ulceration is strongly associated with H. pylori infection. The prevalence of the infection in developed nations rises with age and in the UK approximately 50% of people over the age of 50 years are infected .

In the developing world infection is more common, affecting up to 90% of adults .These infections are probably acquired in childhood by person-to-person contact .The vast majority of colonised people remain healthy and asymptomatic, and only a minority develop clinical disease .

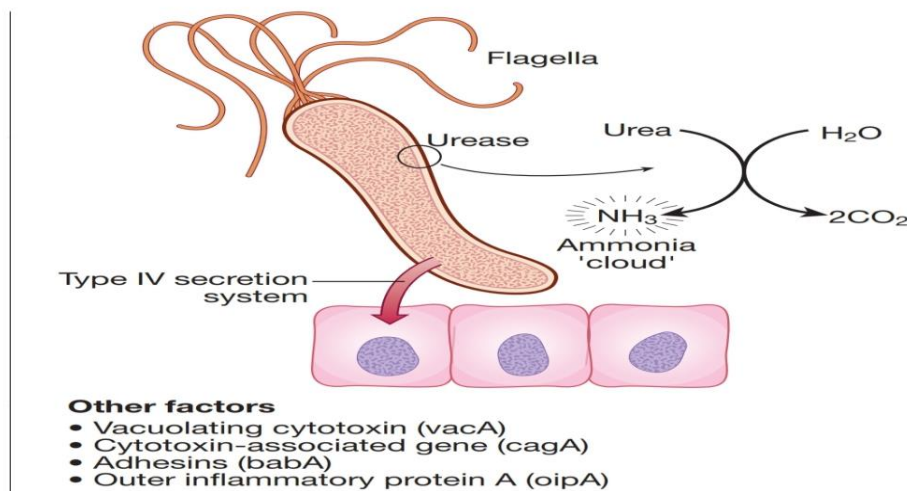
- Around 90% of duodenal ulcer patients and 70% of gastric ulcer patients are infected with H. pylori. The remaining 30% of gastric ulcers are caused by NSAIDs and this proportion is increasing in Western countries as a result of H. pylori eradication strategies.



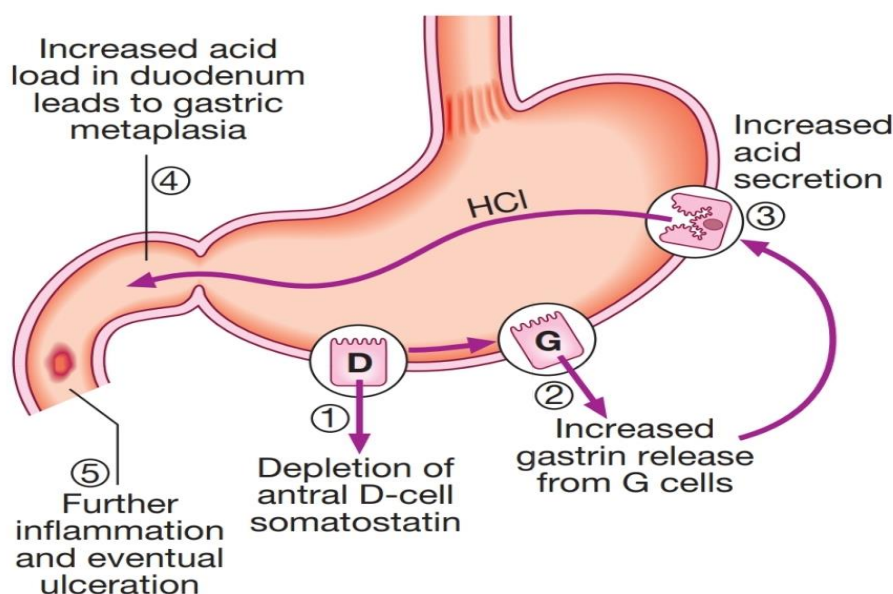
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- *H. pylori* is Gram-negative and spiral, and has multiple flagella at one end , which make it motile, allowing it to burrow and live beneath the mucus layer adherent to the epithelial surface.

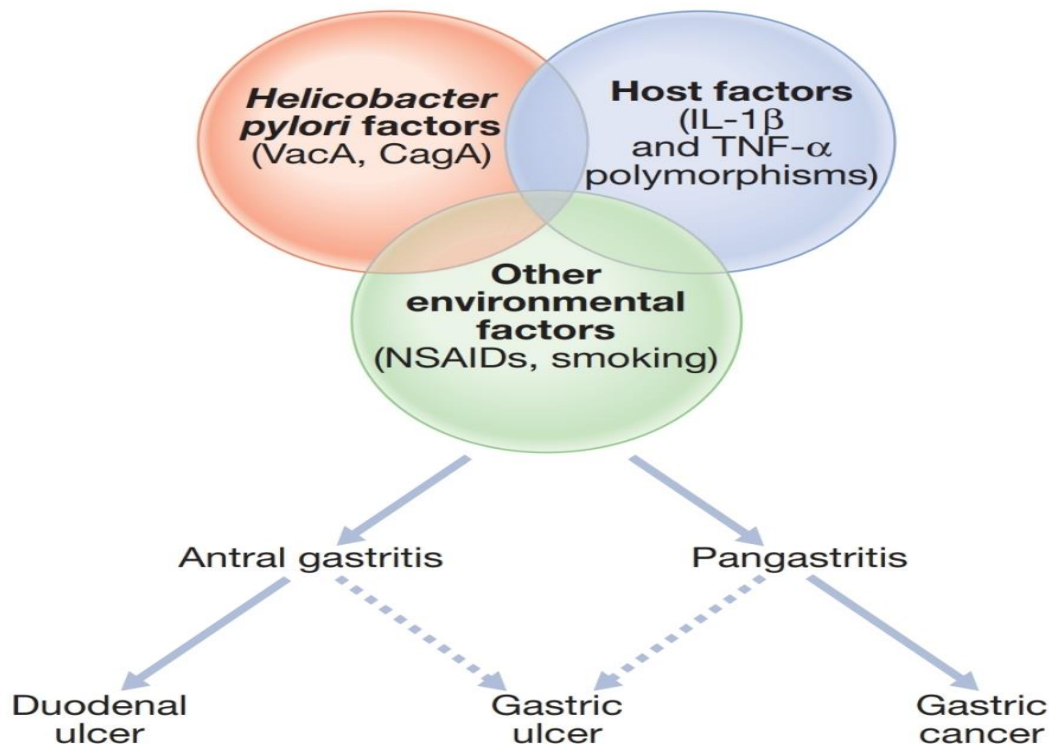


- The organism's produce the enzyme urease. which produces ammonia from urea and raises the pH around the bacterium and between its two cell membrane layers .
- *H. pylori* exclusively colonises gastric-type epithelium and is found in the duodenum only in association with patches of gastric metaplasia. It causes chronic gastritis by provoking a local inflammatory response in the underlying epithelium.
- The distribution and severity of *H. pylori*–induced gastritis determine the clinical outcome. In most people, *H. pylori* causes a mild pangastritis with little effect on acid secretion and the majority develop no significant clinical outcomes .
- In a minority (up to 10% in the West), the infection causes an antral predominant pattern of gastritis characterized by hypergastrinemia and a very exaggerated acid production by parietal cells, which could lead to duodenal ulceration.





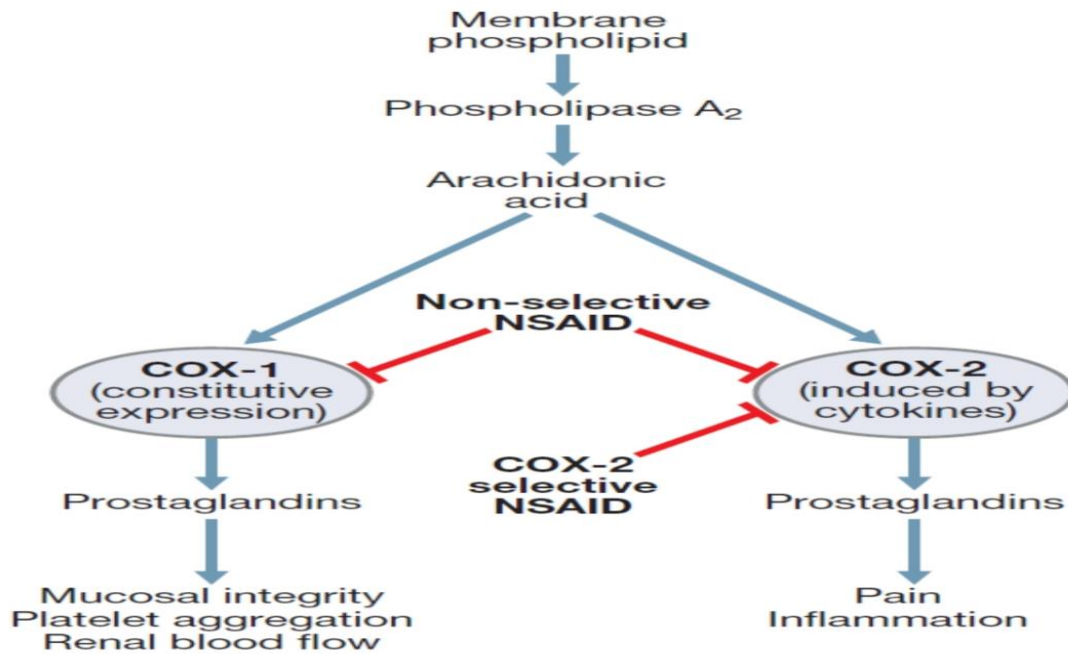
- In a much smaller number of infected people, *H. pylori* causes a corpus predominant pattern of gastritis leading to gastric atrophy and hypochlorhydria.
- This phenotype is much more common in Asian countries, particularly Japan, China and Korea. The hypochlorhydria allows other bacteria to proliferate within the stomach; these other bacteria continue to drive the chronic inflammation and produce mutagenic nitrites from dietary nitrates , predisposing to the development of gastric cancer.
- The effects of *H. pylori* are more complex in gastric ulcer patients compared to those with duodenal ulcers. The ulcer probably arises because of impaired mucosal defence resulting from a combination of *H. pylori* infection, NSAIDs and smoking, rather than excess acid



Pathophysiology : NSAIDs and Smoking

NSAID : Treatment with NSAIDs is associated with peptic ulcers due to impairment of mucosal defenses.

Smoking: Smoking confers an increased risk of gastric ulcer and, to a lesser extent , duodenal ulcer. Once the ulcer has formed, it is more likely to cause complications and less likely to heal if the patient continues to smoke.



Clinical features of PUD

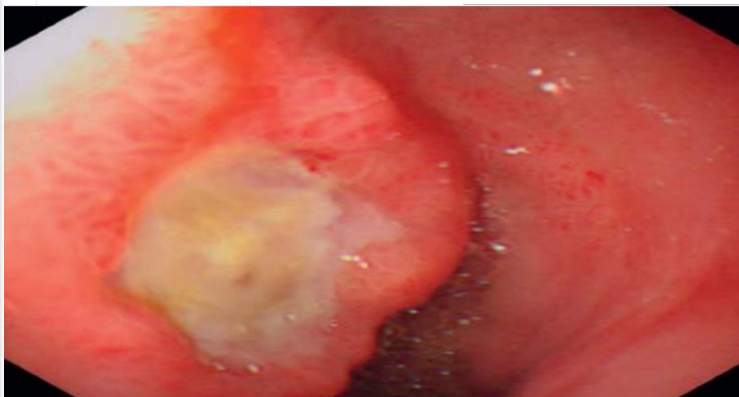
- Peptic ulcer disease is a chronic condition with spontaneous relapses and remissions lasting for decades, if not for life .
- The most common presentation is with recurrent abdominal pain that has three notable characteristics: localization to the epigastrium, relationship to food and episodic occurrence .
- Occasional vomiting occurs in about 40% of ulcer subjects; persistent daily vomiting suggests gastric outlet obstruction. In one-third, the history is less characteristic, especially in elderly people or those taking NSAIDs. In this situation, pain may be absent or so slight that it is experienced only as a vague sense of epigastric unease.
- Occasionally, the only symptoms are anorexia and nausea, or early satiety after meals. In some patients, the ulcer is completely ‘silent’, presenting for the first time with anemia from chronic undetected blood loss, as abrupt hematemesis or as acute perforation; in others, there is recurrent acute bleeding without ulcer pain .
- The diagnostic value of individual symptoms for peptic ulcer disease is poor; the history is therefore a poor predictor of the presence of an ulcer.



Investigations of PUD

- Endoscopy is the preferred investigation . Gastric ulcers may occasionally be malignant and therefore must always be biopsied and followed up to ensure healing. Patients should be tested for *H. pylori* infection. The current options available are listed below.
- Some are invasive and require endoscopy; others are non-invasive. They vary in sensitivity and specificity. Breath tests or faecal antigen tests are best because of accuracy, simplicity and non-invasiveness.

21.34 Methods for the diagnosis of <i>Helicobacter pylori</i> infection			21.36 Indications for <i>Helicobacter pylori</i> eradication	
Test	Advantages	Disadvantages	Definite	
Non-invasive			- Peptic ulcer - Extranodal marginal-zone lymphomas of MALT type - Family history of gastric cancer - Previous resection for gastric cancer <i>H. pylori</i>-positive dyspepsia - Long-term NSAID or low-dose aspirin users - Chronic (> 1 year) PPI users - Extragastric disorders: - Unexplained vitamin B₁₂ deficiency* - Idiopathic thrombocytopenic purpura* - Iron deficiency anaemia* (see text)	
Serology	Rapid office kits available	Lacks specificity		
¹³ C-urea breath test	Good for population studies	Cannot differentiate current from past infection		
Faecal antigen test	High sensitivity and specificity	Requires expensive mass spectrometer		
Invasive (antral biopsy)			Not indicated	
Histology	Cheap, specific (>95%)	False negatives	- Gastro-oesophageal reflux disease - Asymptomatic people without gastric cancer risk factors	
Rapid urease test	'Gold standard'	Takes several days to process		
Microbiological culture	Defines antibiotic sensitivity	Sensitivity 85%	*If <i>H. pylori</i>-positive on testing. (MALT = mucosa-associated lymphoid tissue; NSAID = non-steroidal anti-inflammatory drug; PPI = proton pump inhibitor)	
		Slow and laborious		
		Lacks sensitivity		





Management of PUD

- The aims of management are to relieve symptoms, induce healing and prevent recurrence .
- H. pylori eradication is the cornerstone of therapy for peptic ulcers, as this will successfully prevent relapse and eliminate the need for long-term therapy in the majority of patients

H. pylori eradication

All patients with proven ulcers who are H. pylori-positive should be offered eradication as primary therapy. Treatment is based on a PPI taken simultaneously with two antibiotics (from amoxicillin, clarithromycin and metronidazole) for at least 10 days. High-dose, twice-daily PPI therapy increases efficacy of treatment ,as does extending treatment to 10–14 days. Success is achieved in 80–90% of patients,Resistance to amoxicillin is rare but rates of metronidazole resistance reach more than 50% in some countries and rates of clarithromycin resistance of 20–40% have recently become common. Where the latter exceed 15%, a quadruple therapy regimen, consisting of omeprazole (or another PPI), bismuth subcitrate ,metronidazole and tetracycline (OBMT) for 10–14 days, is recommended .

Management of PUD

General measures

Cigarette smoking, aspirin and NSAIDs should be avoided.Alcohol in moderation is not harmful and no special dietary advice is required.

Maintenance treatment

Continuous maintenance treatment should not be necessary after successful H . pylori eradication. For the minority who do require it, the lowest effective dose of PPI should be used.